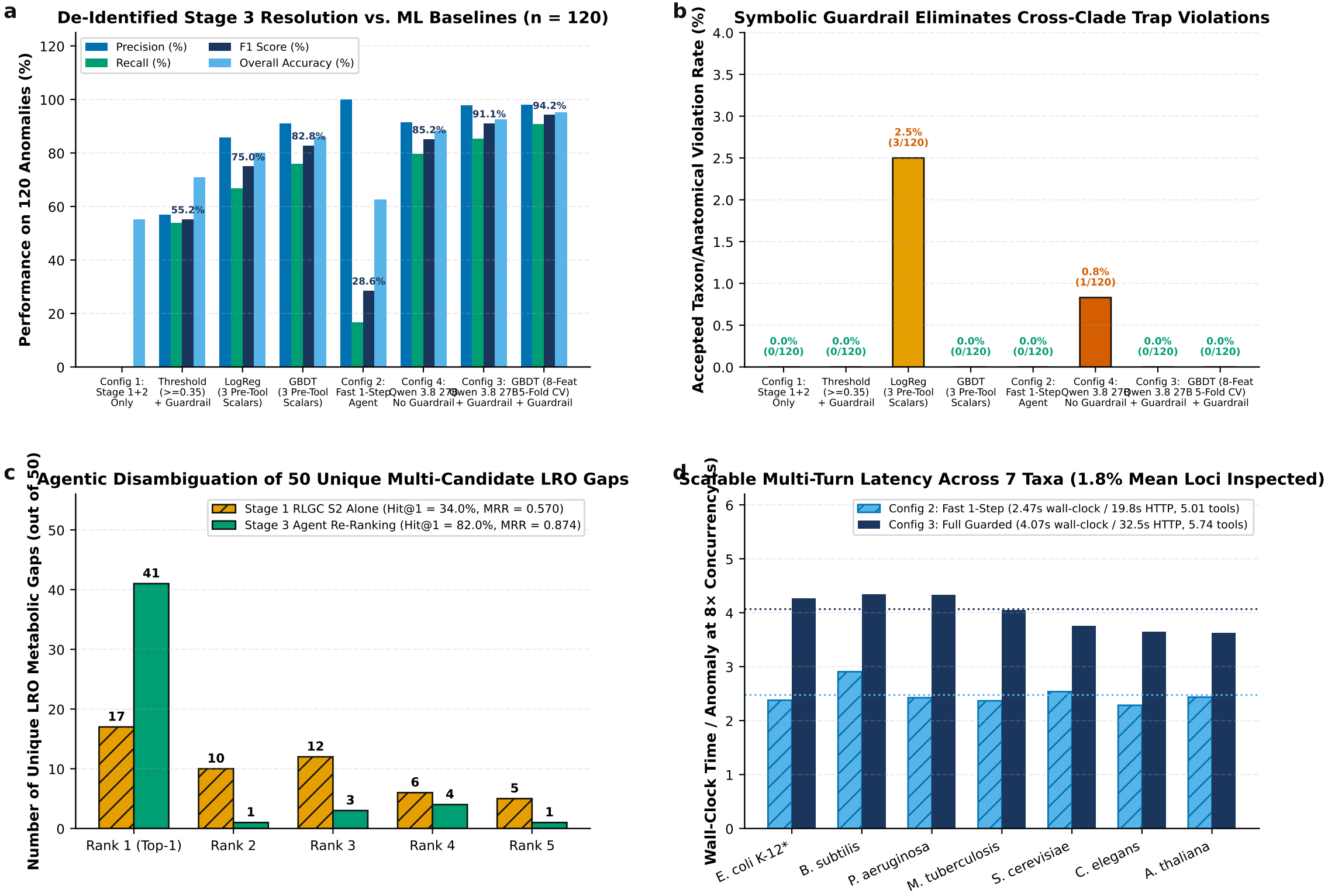

a
De-Identified Stage 3 Resolution vs. ML Baselines (n = 120)
Performance on 120 Anomalies (%)
Precision (%)
Recall (%)
F1 Score (%)
Overall Accuracy (%)
55.2%
75.0%
82.8%
28.6%
85.2%
91.1%
94.2%
Config 1: Stage 1+2 Only
Threshold (>=0.35) + Guardrail
LogReg (3 Pre-Tool Scalars)
GBDT (3 Pre-Tool Scalars)
Config 2: Fast 1-Step Agent
Config 4: Qwen 3.8 27B No Guardrail
Config 3: Qwen 3.8 27B + Guardrail
GBDT (8-Feat 5-Fold CV) + Guardrail
b
Symbolic Guardrail Eliminates Cross-Clade Trap Violations
Accepted Taxon/Anatomical Violation Rate (%)
2.5% (3/120)
0.8% (1/120)
0.0% (0/120)
c
Agentic Disambiguation of 50 Unique Multi-Candidate LRO Gaps
Number of Unique LRO Metabolic Gaps (out of 50)
Stage 1 RLGC S2 Alone (Hit@1 = 34.0%, MRR = 0.570)
Stage 3 Agent Re-Ranking (Hit@1 = 82.0%, MRR = 0.874)
17
41
10
1
12
3
6
4
5
1
Rank 1 (Top-1)
Rank 2
Rank 3
Rank 4
Rank 5
d
Scalable Multi-Turn Latency Across 7 Taxa (1.8% Mean Loci Inspected)
Wall-Clock Time / Anomaly at 8× Concurrency (s)
Config 2: Fast 1-Step (2.47s wall-clock / 19.8s HTTP, 5.01 tools)
Config 3: Full Guarded (4.07s wall-clock / 32.5s HTTP, 5.74 tools)
E. coli K-12*
B. subtilis
P. aeruginosa
M. tuberculosis
S. cerevisiae
C. elegans
A. thaliana